C. Longest Good Array

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: binary search, brute force, math | https://codeforces.com/problemset/problem/2008/C

Today, Sakurako was studying arrays. An array a of length n is considered good if and only if:

- the array a is increasing, meaning a_{i-1} lt a_i for all $2 \le i \le n$;
- the differences between adjacent elements are increasing, meaning $a_i - a_{i-1}$ lt $a_{i+1} - a_i$ for all $2 \le i$ lt n.

Sakurako has come up with boundaries l and r and wants to construct a good array of maximum length, where $l \le a_i \le r$ for all a_i.

Help Sakurako find the maximum length of a good array for the given l and r.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The only line of each test case contains two integers l and r ($1 \le l \le r \le 10^9$).

Output

For each test case, output a single integer — the length of the longest good array Sakurako can form given l and r.

Example

Input
5 1 2 1 5 2 2 10 20 1 1000000000

Output
2 3 1 5 44721

Note

For l=1 and r=5, one possible array could be (1,2,5). It can be proven that an array of length 4 does not exist for the given l and r.

For l=2 and r=2, the only possible array is (2).

For l=10 and r=20, the only possible array is (10,11,13,16,20).